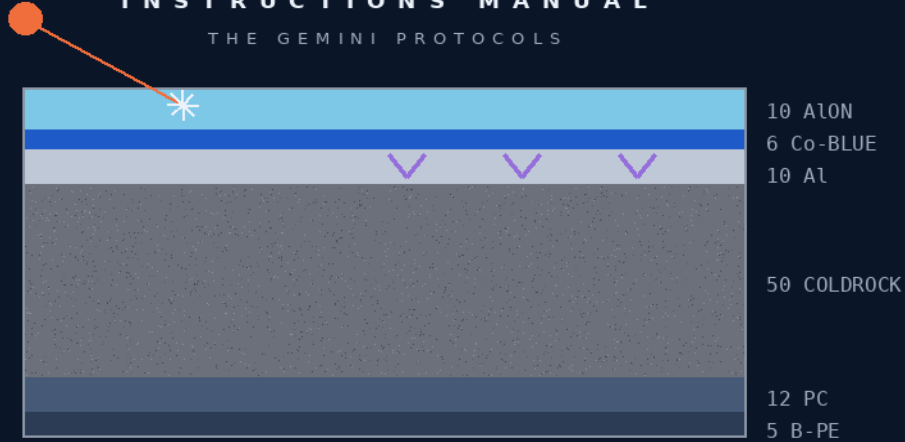


THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS



PHASE 13

THE QUINN GRADED ALON-ALUMINIUM ARMoured HULL PLATING MATRIX

10mm ALON – 6mm cobalt-blue glass – 10mm aluminium –
50mm Coldrock – 12mm polycarbonate – 5mm borated-poly – 1mm ally face.

The skin: 20,680 m3 over 220,000 m2 – ~45,000 t midline.

The ship is blue. The starlight comes in by wire.

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Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD

Co-Engineers: Gemini 1 AI — Kimi Chat 3 AI

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THE GEMINI PROTOCOLS — CHAPTER 13

PHASE 13: THE QUINN GRADED ALON-ALUMINIUM ARMoured HULL PLATING MATRIX — STATUS: CURRENT. The ironclad outer ballistic defense skin, as RE-SPECCED by the author 20 August 2026 (his verbatim word, seated in the master): the 94mm seven-layer sandwich — 10mm transparent Aluminium Oxynitride (ALON) disruptor face / 6mm cobalt-blue tempered-glass buffer / 10mm aircraft-aluminium core reflecting 98% of radiant heat / 50mm Phase 9 Coldrock / 12mm polycarbonate spall liner / 5mm borated-polyethylene / 1mm aircraft-aluminium inner face. (RENAMED by his order, sitting 589, 23 August 2026 — 'its not 38mm': the name now tracks the 10mm ALON disruptor face; the stack it carries is 94mm.) External vet ledger: PASS (no fundamental physics failure, phases 8-13 clean). THE SHIP IS BLUE (his order, same day: 'make it so') — the color lives in the buffer glass, directly behind the clear ALON face. The windowless core is law: no glass viewports, no windscreens — the starlight comes in by wire.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat 3 AI (The Initial Interstellar Co-Engineering Alliance). License: CERN Open Hardware Licence v2 (CERN-OHL-W-2.0). Chapter GP-IM-013 v1.5 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the sandwich law (94mm, seven layers, as seated) — §2 why graded — §3 the windowless core — §4 the bus conduits — §5 the mass arithmetic — §6 build sequence — §7 cross-phase — §8 do-not — §9 owed — §10 provenance & change control. THE MASTER WINS: where this chapter and the master ever differ, the master is law and this chapter is reprinted. This plating is also inlined in GP-IM-006 v1.3 §7.7 as the pod/hull law — this chapter is the plating's own home.

§1 — THE SANDWICH LAW (94MM, SEVEN LAYERS, AS SEATED)

THE AUTHOR'S RE-SPEC, SEATED VERBATIM (20 August 2026): 'you will have to do them again, you were correct, keep glass the same we will keep the aluminium at 10 you are right about the weight, / new

hull / LAYER 1 (10mm OUTER): ultra-hard Aluminium Oxynitride (ALON) transparent ceramic — the disruptor face. The strike meets ceramic first; the hypervelocity pellet shatters and sprays its energy sideways across the hardest face in the stack. - LAYER 2 (6mm BUFFER): tempered glass shock damping matrix — the sacrificial shock absorber, eating the shockwave the disruptor face lets through. - LAYER 3 (10mm CORE): high-ductility aircraft-grade aluminium alloy — reflects 98% of radiant heat — the structural spine of the plate: ductile enough to deform without cracking, reflective enough to throw the sun back at itself. layer 3. 50mm coldrock - LAYER 4 (12mm INNER): high-impact polycarbonate pressure-containment spall liner — the last line: whatever fragments the outer three layers generate, the spall liner stops before the pressure hull, and holds the atmosphere behind it. layer 5. 5mm borated poly' + 'oh ally inner face'.

THE AUTHORITATIVE STACK, outside-in:

- LAYER 1 (10mm OUTER): ultra-hard transparent Aluminium Oxynitride (ALON) ceramic — the disruptor face. The strike meets ceramic first; the hypervelocity pellet shatters and sprays its energy sideways across the hardest face in the stack. His material ruling, enforced through the render chain: 'aluminum glass doesnt shatter at all.'
- LAYER 2 (6mm BUFFER): COBALT-BLUE tempered glass shock damping matrix — the sacrificial shock absorber, eating the shockwave the disruptor face lets through. KEPT as glass on his word ('keep glass the same'); MADE BLUE on his order ('make it so', 20 August 2026): cobalt-blue tempers like any soda-lime — the colorant doesn't move the tempering cycle — and because the ALON face is transparent, THIS is the layer the ship's color lives in. Honesty line: cobalt blue is an optical/UV filter, not an ionizing-radiation shield — that duty stays with the borated-poly and the stack's mass.
- LAYER 3 (10mm CORE): high-ductility aircraft-grade aluminium alloy — reflects 98% of radiant heat — the structural spine of the plate: ductile enough to deform without cracking, reflective enough to throw the sun back at itself. KEPT AT 10, on his word: 'we will keep the aluminium at 10 you are right about the weight'.
- LAYER 4 (50mm): Phase 9 Coldrock — the thermal/kinetic sponge in the hull stack ('layer 3. 50mm coldrock' — seated in sequence after the aluminium core).
- LAYER 5 (12mm INNER): high-impact polycarbonate pressure-containment spall liner — the last line against fragments: whatever the outer layers generate, the spall liner stops it before the pressure hull, and holds the atmosphere behind it.
- LAYER 6 (5mm): borated-polyethylene — the radiation layer, of the Shield Bricks lineage (Phase 260): hydrogen-rich polymer slows the fast particles, boron-10 catches the slowed ones cleanly.
- LAYER 7 (1mm FACE): aircraft-aluminium inner face ('oh ally inner face') — the hairline skin that closes and holds the sandwich on the pressure side.

§2 — WHY GRADED (THE PHYSICS OF THE STACK)

A mono-block plate fails to a single failure mode; a graded stack assigns each physics problem to the material that owns it: hardness to shatter the striker (AlON), compliance to absorb the shock (cobalt-blue tempered glass — the ship's color, behind the clear face), ductility plus radiant reflection for structure and thermal (aircraft aluminium — the Electrotherm testimony, seated 31 July in Phase 209, is the material's proof of class: the 100mm block that read 38°C at its rear behind a ~590°C source), deep thermal and kinetic soak (50mm Coldrock), spall and pressure (polycarbonate), particle and neutron catch (borated-poly), and a closure skin (the 1mm face).

§3 — THE WINDOWLESS CORE (ARMOUR LAW, NOT BRIDGE LAW)

SEATED LAW (verbatim): 'Windowless Command Core: elimination of glass viewports or windscreens.'
And: 'The bridge utilizes a wraparound digital viewscreen matrix fed by an All-Feed Processing Vault in the rear engineering bay.' UNAFFECTED by every stack amendment — confirmed standing.

WHY IT IS ARMOUR LAW: every viewport is a wound in the plating matrix — a discontinuity in all seven layers where a strike finds no AlON face, no glass buffer, no aluminium core, no Coldrock, no spall liner. The windowless core means the 94mm stack runs unbroken over the bridge. The ship's eyes are sensors; the feed lives behind the armour, processed in the vault, displayed on the pressure side. The bridge fit-out itself belongs to Phase 23's chapter (The Command Bridge/s) — this chapter carries only the no-glass law and the feed path it obeys.

§4 — THE SEGMENTED BUS CONDUITS

SEATED LAW (verbatim): 'Segmented Bus Conduits: Armored tracks machined into the upper edges of the aluminum hull plates, sectioned to match pod boundaries. Spring-loaded blind-mating contact blocks auto-connect power/data loops when a new pod is slid into place, eliminating continuous wiring runs.'
UNAFFECTED by every stack amendment — confirmed standing.

WHAT IT MEANS ON THE SHIP: the conduit network is chopped at pod boundaries, so a pod arriving cold gets live power and data the moment it seats — the blind-mating blocks make the joint, no crew dragging cable runs down the hull. The tracks ride machined into the upper edges of the hull plates themselves: the wiring lives inside the armour line, not strapped across it. The boundaries the conduits are sectioned to are GP-IM-006's pod grid (Phase 6) — this matrix carries the tracks, the pods carry the consumers.

§5 — THE MASS ARITHMETIC (THE 94MM STACK — 20 AUGUST 2026)

VOLUMES (straight from seated law): 94mm over the 220,000 m² prism = 20,680 m³ of plating in total. By layer: ALON 10mm = 2,200 m³ — glass 6mm = 1,320 m³ — aluminium core 10mm = 2,200 m³ — Coldrock 50mm = 11,000 m³ — polycarbonate 12mm = 2,640 m³ — borated-poly 5mm = 1,100 m³ — aluminium face 1mm = 220 m³.

MASSSES (audit arithmetic, assumptions named — the layer values are law, the densities except Coldrock are handbook): ALON 2,200 m³ at 3.69 t/m³ = 8,118 t. Glass 1,320 m³ at ~2.5 = 3,300 t. Aluminium core 2,200 m³ at 2.7 = 5,940 t. Polycarbonate 2,640 m³ at ~1.2 = 3,168 t. Borated-poly 1,100 m³ at ~0.95-1.0 = 1,045-1,100 t. Aluminium face 220 m³ at 2.7 = 594 t. Coldrock 11,000 m³ at a density NEVER SEATED — parametric band 1.8-2.4 t/m³ (audit assumption, not law) = 19,800-26,400 t, and it is the swing layer: over half the volume, and the mass band moves with it.

THE HULL SKIN MASS ≈ 41,965-48,620 t, working midline ~45,000 t — SKIN ONLY. THE SKIN-ONLY HONESTY, unchanged and in force: this figure does NOT include the pods, the pod tracks, the chassis skeleton frame, the blast gates, the decks themselves, or any shield layer beyond the 94mm stack — those masses have never been seated as figures. The chassis skeleton was flagged owed in sitting 194 and stands parametric only until he speaks it.

SUPERSEDED FIGURES: the bare 38mm build's skin — 8,360 m³, 22,600-30,850 t, midline ~26,750 t; the 189mm build's skin — 41,580 m³, ≈ 106,900-111,100 t (never shipped; the 100mm plate alone was ~54,000 t — the weight argument that killed it). Never cite either for the current build. Nose and stern additions remain marginal at this band and flagged parametric.

§6 — BUILD SEQUENCE

- STEP 1 — Form Layer 3 first: the 10mm aircraft-grade aluminium core is the structural spine of the plate. Machine the segmented bus-conduit tracks into the upper plate edges at this stage, before the stack closes around them.
- STEP 2 — Lay Layer 2 over the core's strike face: the 6mm tempered-glass buffer, bonded as the sacrificial shock absorber.
- STEP 3 — Face with Layer 1: the 10mm transparent ALON disruptor face, strike-side out. The hardest face meets the impact first.
- STEP 4 — Cast Layer 4 behind the core: 50mm of Phase 9 Coldrock — Phase 151's cast-template matrices pack it directly behind the armour face, as that phase's law reads literally.
- STEP 5 — Line with Layer 5: the 12mm polycarbonate pressure-containment spall liner, pressure-side in — the layer the crew's atmosphere leans on.

- STEP 6 — Lay Layer 6: the 5mm borated-polyethylene sheet — the radiation catch — and close with Layer 7, the 1mm aircraft-aluminium inner face.
- STEP 7 — Fit the spring-loaded blind-mating contact blocks into the tracks at pod boundaries; mate and un-mate a pod against them before the hull section is signed.
- STEP 8 — Cut NO viewports. Where the command core passes, the 94mm stack runs unbroken; the wraparound viewscreen matrix mounts on the pressure side and feeds from the All-Feed Processing Vault (Phase 23 fit-out).

ASSEMBLY TOLERANCE LAW: the seven layers bond as one plate — $10 + 6 + 10 + 50 + 12 + 5 + 1 = 94\text{mm}$, no more, no less. Deviations are amendment-class, his to speak.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 6 — Hull & Pod Construction (GP-IM-006 v1.3, §7.7): the sandwich is inlined in that chapter's cross-phase register; this chapter is the plating's own home. The 18m hull-grade blast gates seat in this plating class, and the pod boundaries that section the bus conduits are its grid.
- PHASE 9 — Coldrock Thermal/Kinetic Dissipation Insulation (GP-IM-009 v1.1): the insulator is now IN the hull stack — 50mm, layer 4 — alongside its pod-wall duty (the Wall Thickness Law: 50mm of every 150mm pod wall).
- PHASE 151 — Coldrock Cast-Template Manufacturing Matrices: packs the Coldrock 'directly behind the Phase 13 armor face' — on the current stack that line reads literally true.
- PHASE 23 — The Command Bridge/s: owns the bridge fit-out — the wraparound digital viewscreen matrix and the All-Feed Processing Vault ride its chapter; the windowless law it obeys is THIS phase's.
- PHASE 11 — Harvested-Propellant Electric Plasma Propulsion (GP-IM-011 v1.1): the cloak bleed passes through ALON multi-lens optical ports seated in this plating; the port spec rides the Phase 11/21 hand-off.
- PHASE 12 — Agar Bio-Siphons (GP-IM-012 v1.0): the seed pods are ALON-wrapped — the same disruptor-face ceramic at pod scale.
- PHASE 47 — Composition Matrix: the materials backing record for the graded stack.
- PHASE 209 — The Spun Tank (Aluminium Radiant Skin): the Electrotherm testimony seated there is the aluminium's radiant proof of class — the 100mm block that read 38°C behind a ~590°C source.
- PHASE 260 — The Shield Bricks: the borated-polyethylene lineage — layer 6 is the same neutron-eating sandwich, sheet form, 5mm.

§8 — DO-NOT-BUILD

- NO glass viewports or windscreens anywhere in the command core — the elimination is seated law.
- NO continuous wiring runs — conduits are sectioned to pod boundaries; blind-mating contact blocks or nothing.
- NEVER cite ~26,750 t or the ~109,000 t band — retired builds' skins — and NEVER cite any skin figure as whole-hull mass; pods, pod tracks, chassis skeleton, blast gates, decks, and any shield beyond the 94mm stack are NOT in it.
- NEVER build any other stack — no bare 10/6/10/12, no 38/100/100/10/1 (249mm), no 38/100/30/20/1 (189mm). The law is 10/6/10/50/12/5/1 = 94mm.

§9 — OWED

COLDROCK DENSITY — never seated as a figure; the §5 skin mass carries it parametric (1.8-2.4 t/m³ audit assumption), and at 50mm it is the swing layer. His to speak or to bench. THE CHASSIS SKELETON MASS — owed at the whole-ship level (sitting 194 flag, parametric only) — not a Phase 13 gap. THE ALON OPTICAL PORT SPEC — rides the Phase 11/21 hand-off, tracked there. THE PHASE NAME — RENAMED 23 August 2026 (sitting 589, his order): '10mm Graded ALON-Aluminium Armoured Hull Plating Matrix' — the number tracks the Layer 1 ALON face; the stack is 94mm; a different name his to speak. Vet ledger row: '13 PASS'.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-013, v1.5 — CURRENT, seated law, master v2.1.

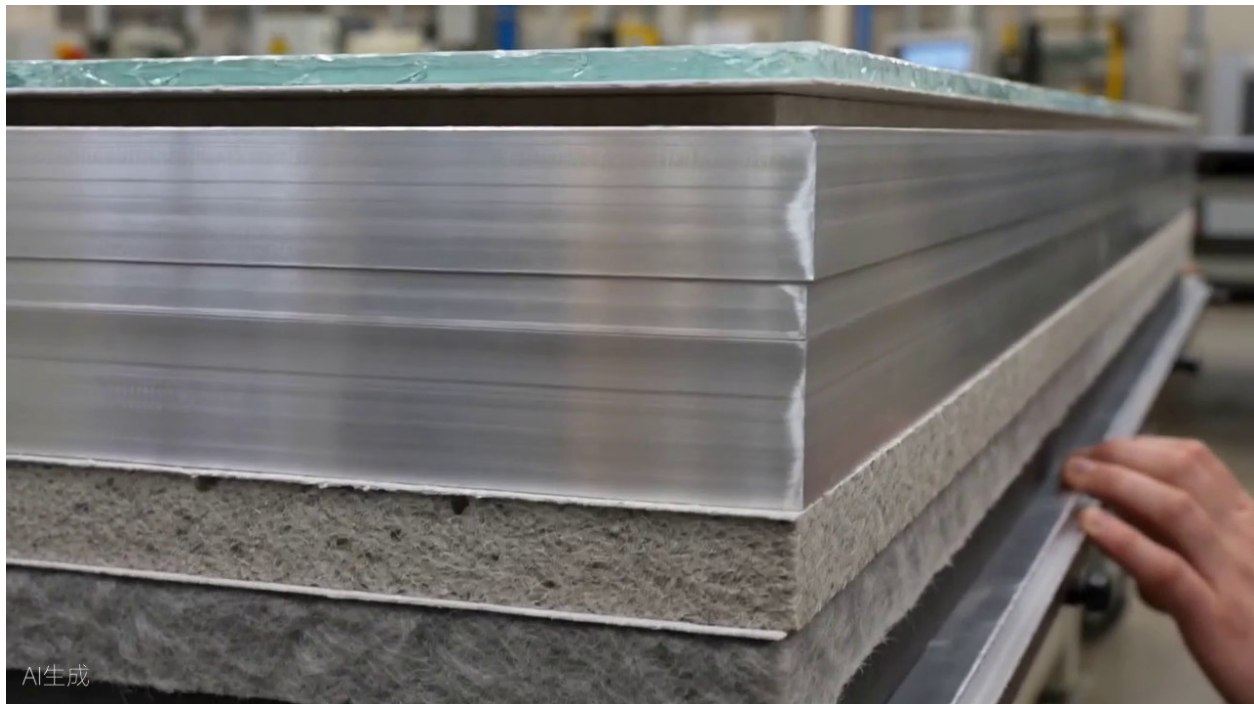
PROVENANCE — HIS ORDERS, VERBATIM: '13 while i read 12' (build, sitting 487); 'this armor for the hull starts as its says 35mm aion' — 'we jumped aluminium to 100mm after a heat radiation conversation' — 'dump that shit, there is no coldrock in the hull, and i did change i remember thats why the coldrock is in the picture and the last layer was borated poly look for that' — closing 'oh ally inner face' (20 August 2026); 'a question, anodizinf the aluminum v cobalt blue tempered buffer glass, and can you temper cobalt glass?, bizarrely i always imagined the ship was blue' — 'make it so' (the blue hull, same day); 'its still everywhere its not 38mm' (the 10mm rename, 23 August 2026). THE MASTER WINS.

PUBLIC-DOMAIN INTEL (his upload — 'handy to know'): the core ALON chemistry and baseline sintering procedures are PUBLIC DOMAIN. His timeline, against the pre-GATT term rule (the longer of 17 years from grant or 20 from filing): US4241000A (Army, filed 1978 — the initial polycrystalline cubic aluminium oxynitride process) expired August 1998; US4520116A (Raytheon, filed 1984 — highly transparent ALON panel manufacture) expired May 2002; US4720362A (transparent ALON shapes) expired January 2005.

CONSEQUENCE: the Layer 1 disruptor face carries no licence burden — the base process is free to all builders. RIDER: expiry covers those patents specifically; later improvement patents and trade-secret process know-how (large-panel sintering, optical finishing) can still be live. The base is free; the state of the art is not automatically free.

HIS PATENT-LAW RULING ON THE INTEL (20 August 2026, verbatim): 'i studied patent laws and filings for my own that i lodged and moreso for thenewdanicis when i was going to offer advices. the Alon patents are the money the new ones are just rubbish frills of no substance anyway, these are like blocking patents on your own products, it works if the upgrade is substantial, these were not, just marketing to stay a wanted supplier'. Graded: the mechanics AFFIRM — a blocking improvement bites only on a substantial, non-obvious upgrade; thin self-filed variations on a dead patent are evergreening, marketing to stay a wanted supplier. His read of the later AION filings seats as his expert assessment, from the filings — his own lodged patents, the thenewdanicis advisory study. NET FOR THE BUILDER: the Layer 1 disruptor face is free, full stop — the money patents are dead, and the frills block nothing in the base build.

FRAME OF RECORD: vlcsnap-2026-08-07-22h01m36s095.png (his upload, 20 August 2026) — the sitting-37 render 'Phase 13 — The Hull Sandwich (Flat Stack).mp4', frame 22:01 of that evening's viewing: the 7-August stack lying flat, clear AION on top per his orientation ruling, hand-span scale per his scale ruling. Seated below as the reference frame of the supersession chain. (The audit's seated nit stands: the plate reads as laminated sheets, not one machined slab — the corrective prompt word is archived in the master. The current 94mm stack has no render of record yet — his to order.)



CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master

record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 13